# Advanced Data Management

Lab 5 — Optimization: estimates, statistics, and bad plans

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## Goals

Learn to read the estimate-vs-actual gap, inspect the statistics behind it, compute a selectivity by hand, and reproduce (then fix) the two classic causes of bad plans: stale statistics and correlated columns. Uses `adm_lab`.

## 1. Estimated vs. actual

```
EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM orders WHERE amount > 90;
EXPLAIN (ANALYZE, BUFFERS)
  SELECT * FROM customers c JOIN orders o ON o.cust_id=c.id WHERE c.
     country='MA';
```

For each node, record `rows=` vs `actual rows`.

## 2. Read the statistics

```
SELECT attname, n_distinct, null_frac, most_common_vals
FROM pg_stats WHERE tablename = 'orders';
```

## 3. Compute a selectivity by hand

```
-- real selectivity of country='MA'
SELECT count(*) FILTER (WHERE country='MA')::float / count(*) FROM
   customers;
-- planner estimate
EXPLAIN SELECT * FROM customers WHERE country = 'MA';
```

Compare your computed selectivity × row count against the planner's estimate.

## 4. Stale statistics

```
INSERT INTO orders (cust_id, product_id, amount, order_date)
SELECT 1+floor(random()*10000)::int, 1+floor(random()*200)::int,
       round((random()*100+5)::numeric,2),
       date '2025-01-01' + (floor(random()*365)::int)
FROM generate_series(1,300000);
```

```
EXPLAIN ANALYZE SELECT * FROM orders WHERE amount > 90;    -- before
    ANALYZE
ANALYZE orders;
EXPLAIN ANALYZE SELECT * FROM orders WHERE amount > 90;    -- after
    ANALYZE
```

## 5. Correlated columns

```
ALTER TABLE customers ADD COLUMN city text;
UPDATE customers SET city = CASE WHEN country='MA' THEN 'Casablanca'
    ELSE 'Other' END;
ANALYZE customers;

EXPLAIN ANALYZE SELECT * FROM customers WHERE country='MA' AND city='
    Casablanca';
CREATE STATISTICS s_geo (dependencies) ON country, city FROM customers;
ANALYZE customers;
EXPLAIN ANALYZE SELECT * FROM customers WHERE country='MA' AND city='
    Casablanca';
```

## 6. Exercises to submit

1. For two queries, report the estimated vs. actual rows at each node and say whether the statistics look accurate.
2. Compute the selectivity of `country='MA'` by hand and compare with the planner's estimated row count. Do they agree?
3. Reproduce the stale-statistics effect: show the estimate before `ANALYZE` and after, and explain why it changed.
4. Reproduce the correlated-columns under-estimate for `country AND city`, then show that `CREATE STATISTICS` improves it. Report both estimates.
5. Short answer: why does a wrong estimate near the leaves of a plan matter more than one near the top?

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### Deliverable

A Jupyter notebook `adm_lab5_<yourname>.ipynb` with all plans and outputs, your by-hand selectivity calculation, the before/after of the stale-stats and correlated-columns experiments, and answers to the five exercises. Attach the `.sql` you ran.

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## Troubleshooting

- Estimates already accurate before `ANALYZE`: insert more rows to make the drift obvious.

- `CREATE STATISTICS` shows little change: make `city` more strongly dependent on `country` (as in Part 5).